Example 1: Binarizer on a NumPy Array

```
from sklearn.preprocessing import Binarizer
import numpy as np

# Sample data
X = np.array([
    [1.2, 3.4, 0.5],
    [2.5, 0.2, 5.1],
    [4.0, 1.5, 2.2]
])

print("Original Data:")
print(X)

# Create Binarizer with threshold = 2.0
binarizer = Binarizer(threshold=2.0)

# Transform data
X_binary = binarizer.fit_transform(X)

print("\nBinarized Data:")
print(X_binary)
```

Output:

```
Original Data:
[[1.2 3.4 0.5]
 [2.5 0.2 5.1]
 [4.  1.5 2.2]]

Binarized Data:
[[0. 1. 0.]
 [1. 0. 1.]
 [1. 0. 1.]]
```

Example 2: Binarizer with a Pandas DataFrame

```
import pandas as pd
from sklearn.preprocessing import Binarizer

# Sample DataFrame
df = pd.DataFrame({
    "Age": [18, 25, 30, 40, 22],
    "Salary": [15000, 30000, 50000, 70000, 20000]
})

print("Original Data:")
print(df)

# Apply Binarizer (threshold = 25000)
binarizer = Binarizer(threshold=25000)
```

```
# Binarize Salary column
df["Salary_Binary"] = binarizer.fit_transform(df[["Salary"]])

print("\nAfter Binarization:")
print(df)
```

Output:

```
   Age  Salary  Salary_Binary
0   18   15000            0.0
1   25   30000            1.0
2   30   50000            1.0
3   40   70000            1.0
4   22   20000            0.0
```

Example 3: Binarizer in a Machine Learning Preprocessing Pipeline

```
import pandas as pd
from sklearn.preprocessing import Binarizer

# Sample dataset
df = pd.DataFrame({
    "Feature1": [10, 20, 30, 40, 50],
    "Feature2": [5, 15, 25, 35, 45]
})

# Create Binarizer
binarizer = Binarizer(threshold=25)

# Transform features
X_binary = binarizer.fit_transform(df)

# Convert back to DataFrame
binary_df = pd.DataFrame(X_binary, columns=df.columns)

print(binary_df)
```

Output

```
   Feature1  Feature2
0       0.0       0.0
1       0.0       0.0
2       1.0       0.0
3       1.0       1.0
4       1.0       1.0
```

When to use `Binarizer`

- Convert continuous values into binary (0/1).
- Create binary features for machine learning models.

- Threshold sensor readings, scores, or probabilities.

Syntax:

```
from sklearn.preprocessing import Binarizer

binarizer = Binarizer(threshold=2.5)
X_binary = binarizer.fit_transform(X)
```

Rule:

- Value **≤ threshold** → `0`
- Value **> threshold** → `1`